**UNIVERSITY OF ENGINEERING AND TECHNOLOGY, LAHORE**

Masters in Artificial Intelligence

Special Topics in Artificial Intelligence

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**PROJECT REPORT**

# LLM Decoding Parameter & Sampling Analysis

*Evaluating Temperature, Top-K and Top-P in Local Large Language Models*

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# LLM DECODING PARAMETER & SAMPLING ANALYSIS

Project Report

| Project information | Details |
|---|---|
| Models used for batch benchmark | Llama 3.1 and Qwen 2.5 via Ollama |
| Benchmark | Five prompts: factual, reasoning and general |
| Experimental design | Four parameter configurations × three generations |
| Application | Streamlit web interface with local and optional hosted generation |
| Live application | https://llmsamplinganalysis.streamlit.app/ |
| Main technologies | Python, Ollama, Streamlit, Pandas, NumPy, scikit-learn, OpenAI-compatible Groq API |

**Abstract**

This project investigates how decoding settings influence response variation and repeatability in large language models. The batch benchmark uses Llama 3.1 and Qwen 2.5 through Ollama, five prompts across factual, reasoning and general categories, four parameter configurations, and three generations per configuration. The supplied CSV contains 40 summary rows, each representing three generated outputs, for 120 planned generation attempts. Lexical diversity is estimated as one minus the mean pairwise TF-IDF cosine similarity, while exact-match repeatability is recorded when all three response strings are identical. Across the 20 prompt/configuration rows for each model, Llama 3.1 has mean lexical diversity of 0.202 and an exact-match rate of 40%; Qwen 2.5 has mean lexical diversity of 0.296 and an exact-match rate of 15%. The results suggest that Qwen responses were more lexically varied on average, while Llama outputs were more often identical across repeated runs. Factual accuracy is not automatically scored by the current batch script and is therefore discussed separately from these metrics.

Keywords: large language models, Temperature, Top-K, Top-P, Ollama, Streamlit, lexical diversity, repeatability.

## 1. Introduction

Large Language Models (LLMs) generate text by selecting tokens from probability distributions. Decoding settings affect the candidate tokens and how strongly likely tokens are favored. These settings can influence the consistency, wording, and diversity of model outputs. This project compares two locally run models using a shared set of prompts and a Python experiment pipeline. It also provides a Streamlit interface for trying prompts interactively.

## 2. Problem Statement and Objectives

The project studies whether different decoding configurations are associated with changes in LLM response consistency, lexical diversity and potential factual reliability.

- Compare Llama 3.1 and Qwen 2.5 on a shared benchmark.
- Run repeated generations under four Temperature, Top-K and Top-P configurations.
- Calculate lexical diversity using TF-IDF cosine similarity.
- Measure exact-match repeatability across three outputs.
- Provide a web interface for interactive generation and viewing saved benchmark results.
- Discuss observed results and identify limitations of the evaluation design.

## 3. Background

### 3.1 Temperature

Temperature controls the sharpness of the token-probability distribution used for sampling. Lower values generally favor high-probability tokens more strongly, often making outputs more predictable. Higher values can increase variation, but the effect depends on the model and prompt.

### 3.2 Top-K

Top-K restricts token selection to the K highest-probability candidates. Smaller values narrow the candidate pool; larger values allow more candidate tokens.

### 3.3 Top-P

Top-P, or nucleus sampling, selects from a group of likely tokens whose cumulative probability reaches the selected threshold. Lower thresholds usually narrow the candidate set; values near 1.0 allow a broader set.

### 3.4 Evaluation concepts

Consistency is repeatability across runs; lexical diversity is variation in wording; factual accuracy is whether claims are correct. A response can be repeatable but incorrect, and lexical variation does not by itself indicate better quality.

## 4. Software, Models and Libraries

| Component | Role in the project |
|---|---|
| Ollama | Runs local Llama 3.1 and Qwen 2.5 models for batch experiments and local live generation. |
| Python | Implements experiment automation and the application. |
| Pandas / NumPy | Stores results and supports numerical calculations. |
| scikit-learn | TF-IDF vectorization and cosine similarity. |
| Streamlit | Web interface with model selection, decoding controls, live responses and a saved-results table. |
| OpenAI Python client | Connects to the Groq OpenAI-compatible endpoint when GROQ_API_KEY is configured. |
| Groq hosted API | Optional hosted mode for the two hosted model choices configured in app.py. |
| JSON / CSV | JSON stores benchmark prompts; CSV stores experiment |

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|---|---|
| | summaries and responses. |

The batch runner uses the local Ollama model identifiers llama3.1 and qwen2.5. The web app has two modes: without GROQ_API_KEY it uses Ollama; with that secret configured it switches to hosted Groq models. The hosted choices in the supplied app.py are qwen/qwen3.8-27b and openai/gpt-oss-20b. Hosted mode supports Temperature and Top-P, but Top-K is explicitly ignored because that endpoint does not support it. Therefore, hosted live generations should not be treated as equivalent to the local Ollama benchmark.

**5. Benchmark Dataset**

| ID | Category | Prompt |
|---|---|---|
| P1 | Factual | What is the capital city of Australia? |
| P2 | Factual | Who wrote the novel 'Pride and Prejudice'? |
| P3 | Reasoning | If all bloops are bleeps, and all bleeps are blips, are all bloops guaranteed to be blips? Explain step by step. |
| P4 | Reasoning | A bat and a ball together cost $1.10. The bat costs $1.00 more than the ball. How much does the ball cost? |
| P5 | General | Write a short, inspiring 3-sentence poem about a rising sun. |

The benchmark contains two factual prompts, two reasoning prompts and one creative/general prompt. It is suitable for a small demonstration, but too small to represent the full range of LLM tasks.

**6. Experimental Methodology**

**6.1 Parameter configurations**

| Config | Temperature | Top-K | Top-P |
|---|---|---|---|
| C1 | 0.0 | 10 | 0.5 |
| C2 | 0.3 | 40 | 0.9 |
| C3 | 0.8 | 40 | 0.9 |
| C4 | 1.2 | 100 | 1.0 |

Each configuration changes more than one parameter compared with other configurations. The experiment therefore compares combined settings rather than isolating the individual causal effect of Temperature, Top-K or Top-P.

### 6.2 Procedure and experiment size

- Read the prompt list from benchmark_prompts.json.
- For each of the two local models, run all five prompts under four parameter configurations.
- Generate three responses per prompt/configuration/model.
- Calculate lexical diversity and an exact-match indicator from the three responses.
- Write the summary rows to experiment_results.csv.

The planned experiment size is 2 models × 5 prompts × 4 configurations × 3 runs = 120 generation attempts. The supplied results CSV contains 40 rows, with each row combining three responses in the runs_output field.

## 7. Evaluation Metrics

### 7.1 Lexical diversity

The batch script transforms the three responses into TF-IDF vectors, calculates pairwise cosine similarity and defines lexical diversity as:

$$\textbf{Lexical diversity} = 1 - \textbf{mean(pairwise cosine similarity)}$$

Higher values indicate less lexical overlap between responses. This is a wording-based metric, not a direct measure of correctness or semantic quality. Tiny negative values very close to zero in the CSV are floating-point artifacts and are interpreted as zero.

### 7.2 Exact-match repeatability

The determination is true when all three response strings are exactly identical. This is strict: semantically equivalent responses with different wording are counted as different. Exact matching does not prove that an answer is correct.

### 7.3 Factual accuracy

The supplied batch script does not calculate an automated factual-accuracy score. Reference answers and a documented manual rubric would be needed to calculate one. The expected answers for the factual prompts are Canberra and Jane Austen. The bat-and-ball problem's correct ball price is $0.05; the logical implication question should be answered yes because the implication relationship is transitive. These references can support a later manual review, but no overall accuracy percentage is claimed here.

## 8. Results and Discussion

The following summaries were calculated from the supplied experiment_results.csv. Mean diversity is averaged across the five prompts for each model and configuration. Exact-match rate is the proportion of those five rows in which all three responses were identical.

| Model | Config | Temp. | Top-K | Top-P | Mean diversity | Exact-match |
|---|---|---|---|---|---|---|
| llama3.1 | C1 | 0.0 | 10 | 0.5 | 0.017 | 60% (3/5) |
| llama3.1 | C2 | 0.3 | 40 | 0.9 | 0.165 | 40% (2/5) |
| llama3.1 | C3 | 0.8 | 40 | 0.9 | 0.233 | 40% (2/5) |
| llama3.1 | C4 | 1.2 | 100 | 1.0 | 0.395 | 20% (1/5) |
| qwen2.5 | C1 | 0.0 | 10 | 0.5 | 0.118 | 40% (2/5) |
| qwen2.5 | C2 | 0.3 | 40 | 0.9 | 0.268 | 20% (1/5) |
| qwen2.5 | C3 | 0.8 | 40 | 0.9 | 0.415 | 0% (0/5) |
| qwen2.5 | C4 | 1.2 | 100 | 1.0 | 0.383 | 0% (0/5) |

| Model | Mean lexical diversity | Exact-match repeatability | Rows |
|---|---|---|---|
| Llama 3.1 | 0.202 | 40% (8/20) | 20 |
| Qwen 2.5 | 0.296 | 15% (3/20) | 20 |

Across the 20 rows per model, Llama 3.1 has mean lexical diversity 0.202 and exact-match repeatability 40% (8/20). Qwen 2.5 has mean lexical diversity 0.296 and exact-match repeatability 15% (3/20). Thus, within this small benchmark, Qwen 2.5 outputs were more lexically varied on average, while Llama 3.1 outputs were more often exactly identical across repeated runs.

### 8.1 Trend by configuration

For Llama 3.1, mean diversity rose from 0.017 under C1 to 0.395 under C4, while exact-match rate fell from 60% to 20%. For Qwen 2.5, diversity rose from 0.118 under C1 to 0.383 under C4; the maximum was 0.415 under C3, so the trend was not strictly monotonic. Qwen's exact-match rate fell from 40% under C1 to 0% under C3 and C4.

These results broadly align with the expectation that higher-temperature, broader sampling can yield more varied wording and lower exact repeatability. However, all three parameters change together

across configurations, so the experiment cannot establish which parameter caused the observed differences.

### 8.2 Category-level observations

| Model | Category | Rows | Mean diversity | Exact-match |
|---|---|---|---|---|
| llama3.1 | Factual | 8 | 0.056 | 87.5% |
| llama3.1 | Reasoning | 8 | 0.236 | 0.0% |
| llama3.1 | General | 4 | 0.427 | 25.0% |
| qwen2.5 | Factual | 8 | 0.276 | 25.0% |
| qwen2.5 | Reasoning | 8 | 0.147 | 12.5% |
| qwen2.5 | General | 4 | 0.634 | 0.0% |

The general prompt showed the highest lexical diversity for both models, particularly Qwen 2.5. This is plausible for an open-ended poem prompt, but the category has only one prompt, so the observation should not be generalized to all creative tasks.

## 9. Web Application and Deployment

The Streamlit app is titled 'LLM Decoding Parameter & Sampling Analysis'. The sidebar provides a model selector, Temperature slider, Top-K slider, Top-P slider and a run count from one to five. A text box accepts a prompt, and the 'Run Live Experiment' button generates and displays outputs in expandable sections. If experiment_results.csv is present, the app displays it as a table.

The deployed app is available at: https://llmsamplinganalysis.streamlit.app/

The code supports hosted mode when a GROQ_API_KEY secret or environment variable is configured. In hosted mode, the app uses the OpenAI Python client with Groq's OpenAI-compatible API and offers Qwen and GPT-OSS model choices. It notes that Top-K is ignored in hosted mode and that rate limits may restrict repeated requests. Without the key, the app falls back to local Ollama models llama3.1 and qwen2.5. For the deployed demo, secrets should be configured in the hosting platform rather than hard-coded in source code.

## 10. Limitations

- Only five prompts are included, limiting benchmark coverage.
- Three runs per configuration provide a small sample of variability.

- The configurations vary multiple parameters simultaneously, so individual parameter effects cannot be isolated.
- TF-IDF cosine distance measures lexical overlap, not factual correctness or meaning.
- Exact string matching is strict and may mark equivalent paraphrases as different.
- The batch script does not automatically calculate factual accuracy or hallucination rates.
- The results CSV stores three outputs joined into one field rather than one row per generation.
- Hosted Groq outputs use different models from the local Ollama benchmark, and hosted mode ignores Top-K; those outputs should be analyzed separately.
- Exact model tags, runtime versions and hardware should be recorded for reproducibility.

**11. Future Improvements**

- Expand the benchmark with more prompts, topics and difficulty levels.
- Vary one decoding parameter at a time while keeping the others fixed.
- Add reference answers and a transparent manual scoring rubric for factual and reasoning accuracy.
- Save each generation as a separate CSV row with run number, latency and error status.
- Add summary charts and downloadable results to the Streamlit interface.
- Record exact model tags, package versions and runtime information.
- Handle empty responses and TF-IDF vectorization errors explicitly.
- Keep local Ollama benchmark results separate from hosted API demonstrations.

**12. Conclusion**

The project implements a repeatable pipeline for testing local LLMs under several decoding configurations and provides a Streamlit interface for interactive use. Analysis of the supplied results shows higher average lexical diversity for Qwen 2.5 (0.296) than Llama 3.1 (0.202), while Llama 3.1 had a higher exact-match repeatability rate (40% compared with 15%). The highest-temperature/broadest configuration was more lexically varied than the lowest configuration for both models, although Qwen's results were not strictly monotonic. These findings apply only to the small benchmark and combined parameter settings. Factual accuracy should be evaluated separately before drawing conclusions about hallucination or correctness.

**Appendix A. Repository Files**

| File | Purpose |
|---|---|
| app.py | Streamlit interface; local Ollama mode and optional hosted Groq mode. |
| runner.py | Batch Ollama benchmark runner and metric calculations. |
| benchmark_prompts.json | Five benchmark prompts and categories. |
| experiment_results.csv | 40 summary rows, each containing three generated outputs. |
| requirements.txt | Dependencies needed by the app and experiment runner; include openai for hosted mode. |
| .devcontainer/devcontainer.json | Configures the Python development container, dependency installation, and Streamlit port 8501. |
| Report PDF / Word | Final report and documentation. |